Assignment 5

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6. We continue to consider the use of a logistic regression model to predict the probability of default using income and balance on the Default data set. In particular, we will now compute estimates for the standard errors of the income and balance logistic regression coeffcients in two diferent ways: (1) using the bootstrap, and (2) using the standard formula for computing the standard errors in the glm() function. Do not forget to set a random seed before beginning your analysis.

(a) Using the summary() and glm() functions, determine the estimated standard errors for the coefcients associated with income and balance in a multiple logistic regression model that uses both predictors

```
library(ISLR2)

## Warning: package 'ISLR2' was built under R version 4.2.3

set.seed(221)
fit.mod <- glm(default ~ income + balance, data = Default, family =
"binomial")
summary(fit.mod)$coef[, "Std. Error"]

##  (Intercept)       income      balance
## 4.347564e-01 4.985167e-06 2.273731e-04

glm.se <- summary(fit.mod)$coef[, "Std. Error"]
```

(b) Write a function, boot.fn(), that takes as input the Default data set as well as an index of the observations, and that outputs the coefcient estimates for income and balance in the multiple logistic regression model.

```
boot.fn <- function(data, indices) {
  sample_data <- data[indices, ]
  model <- glm(default ~ income + balance, data = sample_data, family =
"binomial")
  return(coef(model))
}
```

(c) Use the boot() function together with your boot.fn() function to estimate the standard errors of the logistic regression coefcients for income and balance.

```
library(boot)
set.seed(423)

boot.strp <- boot(data = Default, statistic = boot.fn, R = 1000)
boot.se <- apply(boot.strp$t, 2, sd)
```

(d) Comment on the estimated standard errors obtained using the glm() function and using your bootstrap function.

```
print("Standard Errors (glm):")

## [1] "Standard Errors (glm):"

print(summary(fit.mod)$coef[, "Std. Error"])

##  (Intercept)       income      balance
## 4.347564e-01 4.985167e-06 2.273731e-04

print("Standard Errors (bootstrap):")

## [1] "Standard Errors (bootstrap):"

print(boot.se)

## [1] 4.505711e-01 4.926907e-06 2.335848e-04
```

The standard error for the bootstrap function is less compared to glm making it much more precise.

9. We will now consider the Boston housing data set, from the ISLR2 library.

(a) Based on this data set, provide an estimate for the population mean of medv. Call this estimate μˆ.

```
data(Boston)
mu.h <- mean(Boston$medv)
mu.h

## [1] 22.53281
```

(b) Provide an estimate of the standard error of μˆ. Interpret this result. Hint: We can compute the standard error of the sample mean by dividing the sample standard deviation by the square root of the number of observations.

```
se.mu.h <- sd(Boston$medv) / sqrt(length(Boston$medv))
se.mu.h

## [1] 0.4088611
```

Interpretation: The standard error of the sample mean provides a measure of the uncertainty or variability in our estimate of the population mean. A larger standard error suggests more variability and less precision in our estimate.

(c) Now estimate the standard error of μˆ using the bootstrap. How does this compare to your answer from (b)?

```
boot_fn <- function(data, indices) {
  sample_data <- data[indices]
  return(mean(sample_data))
}
```

```r
boot_results <- boot(data = Boston$medv, statistic = boot_fn, R = 1000)

se.boot.mu.h <- sd(boot_results$t)
se.boot.mu.h

## [1] 0.4207522
```

The bootstrap result for se is higher sugesests there is more variability and and less precision of the mu.h.

(d) Based on your bootstrap estimate from (c), provide a 95 % confdence interval for the mean of medv. Compare it to the results obtained using t.test(Boston$medv). Hint: You can approximate a 95 % confdence interval using the formula [ˆμ − 2SE(ˆμ), μˆ + 2SE(ˆμ)].

```r
conf.int.boot <- quantile(boot_results$t, c(0.025, 0.975))
conf.int.boot

##     2.5%    97.5%
## 21.73696 23.37487

t.conf.int <- t.test(Boston$medv)$conf.int
t.conf.int

## [1] 21.72953 23.33608
## attr(,"conf.level")
## [1] 0.95
```

(e) Based on this data set, provide an estimate, μˆmed, for the median value of medv in the population.

```r
med.h <- median(Boston$medv)
med.h

## [1] 21.2
```

(f) We now would like to estimate the standard error of μˆmed. Unfortunately, there is no simple formula for computing the standard error of the median. Instead, estimate the standard error of the median using the bootstrap. Comment on your fndings.

```r
boot_median_fn <- function(data, indices) {
  sample_data <- data[indices]
  return(median(sample_data))
}

boot.med <- boot(data = Boston$medv, statistic = boot_median_fn, R = 1000)
se.boot.med.h <- sd(boot.med$t)
se.boot.med.h

## [1] 0.361759
```

(g) Based on this data set, provide an estimate for the tenth percentile of medv in Boston census tracts. Call this quantity μˆ0.1. (You can use the quantile() function.)

```r
p10.h <- quantile(Boston$medv, 0.1)
p10.h
```

```
##   10%
## 12.75
```

(h) Use the bootstrap to estimate the standard error of μˆ0.1. Comment on your fndings.

```r
boot.p10 <- function(data, indices) {
  sample_data <- data[indices]
  return(quantile(sample_data, 0.1))
}

boot.p10.fin <- boot(data = Boston$medv, statistic = boot.p10, R = 1000)

#
se.boot.p10.h <- sd(boot.p10.fin$t)
se.boot.p10.h
```

```
## [1] 0.4927669
```

There is a moderate level of variability since our result was 0.5003814.